



RefAI Accountability Document

A controllable extraction and triage pipeline for Dutch archaeological news bulletins

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THESIS ACCOUNTABILITY DOCUMENT

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Statement of Originality

I hereby declare that this dissertation is an original piece of work, written by myself alone. Any information and ideas from other sources are acknowledged fully in the text and notes.

Amsterdam, 30 June 2026

A handwritten signature in green ink that reads "T.O.S. Verwoert". The signature is stylized, with a large, sweeping initial 'T' and a horizontal line underneath the name.

Tobias O.S. Verwoert

Preface

This accountability document accompanies the development of RefAI, a prototype pipeline for extracting structured archaeological information from historical KNOB Archaeological News bulletins. It serves both as a methodological account and as a technical accountability record. It explains why particular design choices were made, how the pipeline was tested, where it performs well, where it remains uncertain, and how those uncertainties are kept visible in the outputs and evaluation.

The project grew out of my internship at the beginning of 2026 and developed further in dialogue with Philip Verhagen and Matthias Lang. RefAI explores whether information contained in the bulletins can be mined from text, converted into a structured dataset and later transferred to a geodatabase for their wider research project. It is therefore best understood as a data-mining experiment with an archaeological purpose. The central question is not whether artificial intelligence can produce a perfect archaeological database, but whether a controllable workflow can make difficult PDF material more searchable, structured and reviewable.

I use this document to make those choices explicit. It therefore records both the strengths and limitations of the pipeline. RefAI should be read as a research prototype and triage tool: it can extract a substantial amount of low-hanging structured information, but it also needs review flags, quarantine logic, reference documents, gold-standard comparison and human judgement to remain methodologically responsible.

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1. Project Context

RefAI was developed to transform semi-structured Dutch archaeological news bulletin text into structured, reviewable data. The project grew out of my internship in January 2026, titled Improving a KNOB Bulletin Text-Mining Pipeline for Archaeological Geo-Data, and was developed in collaboration with, and partly in response to, the research needs of Philip Verhagen at Vrije Universiteit Amsterdam and Matthias Lang at the University of Bonn. The source material consists of Archaeological News bulletins, or Archeologisch Nieuws (Archaeological News), published by the KNOB (Koninklijke Nederlandse Oudheidkundige Bond). These bulletins are valuable because they contain many short archaeological notices, but they are difficult to convert into structured data because their layout, typography, column structure, scan quality and editorial conventions vary strongly between years. Similar attempts to make Dutch archaeological literature computationally searchable show both the value of domain-specific information retrieval and the importance of entity-centred access to places, periods and artefacts (Brandesen et al., 2021a, 2021b). Key technical terms used throughout this document are defined in Appendix A.

RefAI was initially developed within an ongoing collaborative research project involving Vrije Universiteit Amsterdam and the University of Bonn. The immediate objective of this wider project is to mine archaeological reports for information that can expand and enrich an existing database of Roman settlements along the Lower Germanic Limes. This specific application shaped RefAI's emphasis on place resolution, dating, archaeological features, finds and standardised archaeological terminology. At the same time, the pipeline was designed as a broader research prototype. Its architecture may also be applied to other periods, regions and collections of Dutch archaeological literature, provided that the reference resources and validation rules are adapted to the new research context.

Reading the text from a PDF is only the first problem. RefAI must then decide what the words mean in their archaeological context and in which output field they belong. The same word can have several meanings. Huizen may refer to the Dutch settlement or simply to multiple houses, while Ring may describe a genuine artefact or appear accidentally through OCR noise or as part of another word. RefAI must therefore extract information, assign it to the correct category, normalise it and indicate doubt where the context is insufficient. Historical-document research shows that OCR errors, changing language and unstable layouts make these decisions particularly difficult (Ehrmann et al., 2021).

The thesis positions RefAI as a controllable hybrid, lookup-first extraction workflow rather than an unconstrained LLM-first system. The hybrid design combines deterministic rules, curated reference data, local language processing and an optional LLM review layer. Models may propose candidates, but gazetteer logic, ABR terminology, evidence from the source text and explicit rulebook decisions constrain what may enter the final output. The aim is to reduce hallucination risk and keep decisions inspectable for archaeologists. This resembles grounded or retrieval-augmented processing, although RefAI relies mainly on deterministic local tables rather than unrestricted generation (Lewis et al., 2020).

This origin matters for the way the project should be evaluated. RefAI was not built as an abstract machine-learning exercise, but as a practical response to a recurring data problem in digital archaeology: valuable archaeological observations are present in published text, but they are not immediately reusable as structured data. The pipeline therefore sits between text mining, data plumbing and archaeological interpretation. It has to make old publication formats computationally accessible while still respecting the fact that archaeological categories are often interpretative.

1.1 Aim and Scope

The aim of RefAI is to design, implement and evaluate a reproducible workflow for extracting archaeological information from historical KNOB bulletin PDFs. The pipeline focuses on place, province, dating or datering (dating or period assignment), feature context or sporen (features/traces), findings or vondsten (finds/artefacts), materials and ABR-compatible labels and codes. It is intended to create a structured first-pass dataset, preserve textual evidence and expose uncertainty for human review. It does not attempt to replace archaeological interpretation.

The scope is deliberately limited to a research prototype and triage tool. The seven selected bulletins cover different years, publication designs, layouts and OCR conditions, but they are not representative of all KNOB publications or Dutch archaeological literature. The same documents also informed iterative development, so the evaluation concerns a partly familiar test environment. Reliability is therefore understood as retrieving useful information consistently enough for professional review, constraining unsupported decisions and making errors inspectable. A further design objective is to keep the computational black box as small and transparent as possible: whenever practical, an output should be traceable to source evidence, a named rule, a reference entry or a recorded model intervention rather than to unexplained automated reasoning.

1.2 Research Questions

The following main question and four sub-questions provide the analytical structure for the design, evaluation and conclusion of this thesis.

Main research question: *How can a controllable, low-hallucination extraction workflow be built for Dutch archaeological texts that reliably extracts place, province, dating, features and objects into reproducible structured data for professional use?*

Sub-question 1 - Segmentation robustness: *How can rule-based section segmentation be made reliable across different KNOB bulletin layouts, and what is the impact of segmentation errors on downstream extraction?*

Sub-question 2 - Hybrid extraction and hallucination: *To what extent can transformer-based candidate detection, combined with deterministic lookup and ABR mapping tables, reduce hallucination compared with an LLM-first pipeline?*

Sub-question 3 - Reference data and maintenance: *Which reference datasets for places, periods, ABR features and objects, dating expressions and materials contribute to accuracy and auditability, and how can a feedback loop maintain and expand them systematically?*

Sub-question 4 - Uncertainty and professional validation: *How should uncertainty, provenance and evidence snippets be represented in the output so that professional users can validate results efficiently without rereading full PDFs?*

These questions are answered through implementation evidence, graded comparison, review and error analysis, and explicit discussion of technical and ethical limitations. No claim is made that the current evaluation establishes general model accuracy. Every additional document may introduce a new column structure, heading convention, homonym, place-feature conflict or vocabulary gap.

Continuing to optimise only against the seven selected documents would risk overfitting the pipeline to known cases. The project therefore stops at a defensible baseline and asks whether RefAI retrieves low-hanging fruit, makes difficult cases visible, and can be reproduced and extended by later researchers.

2. Data and Source Material

The evaluation uses seven selected KNOB test PDFs from different periods in the publication history of the bulletin. They differ not only in year and scan quality, but also in typography, column structure, heading conventions, page design and the way Archaeological News is separated from other editorial sections. The set is small enough for manual inspection and gold-standard correction, yet varied enough to expose recurring failure modes such as multi-column reading order, OCR-damaged text, captions, section transitions, bibliography or museum-news leakage, ambiguous place names and dense archaeological terminology. It is therefore a deliberately varied stress-test sample rather than a statistically representative corpus (Geburu et al., 2021).

Several reference resources support the workflow. Gazetteer and place/province resources support geographical normalisation. ABR period, feature, finding and material resources support standardised archaeological classification. Custom reference data in RefAI_Custom_Aliases.xlsx stores project-specific aliases, bucket rules, ambiguity rules and suppression rules. Moving such decisions into reference documents is methodologically important because it separates domain knowledge from procedural code and makes later correction possible without endlessly adding opaque hardcoded exceptions. Controlled vocabularies are especially important in cultural-heritage data because they reduce terminological variation and support interoperability, while also requiring transparent decisions about preferred concepts (Harpring, 2010; CIDOC CRM Special Interest Group, n.d.).

The reference documents are not passive lookup lists; together they function as a small, local and editable knowledge base. In practical terms they act as the memory available to the SLM-oriented pipeline when no LLM is used. The broader and better curated this knowledge base becomes, the more historical spellings, place names, periods and archaeological terms RefAI can recognise and normalise. This relationship is not unlimited: when the correct concept is missing, or when source text is linked to the wrong entry, the pipeline can still return an empty or incorrect value. The advantage over an opaque generated answer is that the relevant alias, rule or preferred category can be inspected and corrected. Storing these decisions outside the main code also supports the FAIR principles by making the knowledge resources easier to find, interpret and reuse (Wilkinson et al., 2016).

3. Accountability for the Use of AI

Artificial intelligence was used in this thesis in two different ways, and I separate these uses deliberately. On the one hand, AI is part of the research object itself: RefAI includes local language processing, reference-table matching and an optional LLM-based review step. On the other hand, AI tools were also used while I was programming and writing. In both cases, they were support tools rather than independent researchers or sources of archaeological evidence. The research questions, archaeological decisions, interpretation of results and final responsibility for the thesis remained my own.

OpenAI Codex was mainly used while developing and improving the Python pipeline. It was most useful when debugging became difficult, or when larger parts of the notebook, script, reference documents and output structure had to be reorganised. Codex helped me understand errors, compare possible solutions and keep changes technically consistent. I did not treat its suggestions as automatically correct. When a change looked useful, I checked it through backups, syntax checks, targeted tests and new pipeline runs. The practical question was always whether the change improved RefAI across the test documents, or whether it simply fixed one case while creating a new problem elsewhere.

Anthropic Claude was used mostly for writing support. I have dyslexia, and this can make it harder to turn complex thoughts, technical decisions and methodological reasoning into clear academic English. Claude helped me reformulate notes, smooth sentence structure and make arguments easier to follow. I understand this as an accessibility and communication aid. The ideas, observations, evaluations and conclusions came from me, and I checked the wording against the actual pipeline, outputs, gold-

standard files and cited sources. Claude was not used as archaeological evidence and did not decide the interpretation.

This use of AI was also directly connected to the subject of the thesis. RefAI investigates whether AI-assisted methods can be useful in an archaeological extraction workflow, so it would be artificial to pretend that AI played no role in the process. At the same time, the project was not outsourced to AI. Codex was used selectively for technical support, and Claude was used to help express the work more clearly. The research design, review of extracted data, source selection, methodological choices and final judgements remained human decisions.

Suggestions from Codex and Claude were therefore treated as proposals. They still had to be tested, checked and sometimes rejected. This was especially important for code, because a change that fixes one document can easily break another. For writing, AI-assisted formulations were checked against the implemented workflow, the evaluation files and the literature. Quantitative statements were not taken from chat output, but from RefAI results, gold-standard workbooks and comparison scripts. This follows the same basic logic as model cards and datasheets: the origin, purpose and limitations of AI-assisted work should remain visible rather than hidden behind polished output (Gebru et al., 2021; Mitchell et al., 2019).

The ethical position of the project is based on a few practical principles: proportionality, human oversight, traceability and avoiding preventable harm. RefAI is not intended to replace archaeological judgement. It is meant to make difficult PDF material more searchable and reviewable. For that reason, the pipeline preserves evidence snippets, produces audit outputs, uses review flags and keeps uncertain cases visible instead of hiding them behind confident-looking output. These choices connect to wider responsible-AI principles such as transparency, accountability, robustness and meaningful human control (Association for Computing Machinery, 2018; European Commission High-Level Expert Group on Artificial Intelligence, 2019; National Institute of Standards and Technology, 2023; UNESCO, 2021).

Although this thesis does not treat RefAI as a high-risk AI system in the legal sense, its design choices were informed by broader European discussions about responsible and accountable AI. The EU AI Act is relevant here as part of that wider context, because it emphasises risk awareness, transparency, human oversight and accountability in AI systems (European Union, 2024). The optional LLM layer used Anthropic Claude Haiku in the deposited version, specifically `claude-haiku-4-5-20251001`, but only as a targeted second reader for selected review cases. It was not used as the foundation of the extraction method.

Privacy, cost and dependency were also considered. API keys are not stored in the deposited code, and the Anthropic review layer can be switched off. The KNOB bulletins are published archaeological sources rather than confidential documents, but sending text to a commercial API still creates an external dependency. For that reason, the local SLM-oriented route remains important. OCR, layout reconstruction, reference lookup, ABR validation and the core extraction logic can run without a commercial LLM. The final responsibility for this thesis remains mine: AI tools helped with technical problem-solving and clearer communication, but they did not remove the need for archaeological judgement, source criticism or manual checking. Any remaining mistakes in the pipeline, interpretation or written argument are therefore also my responsibility.

4. Methodological Design of the Hybrid Pipeline

RefAI is built as a hybrid pipeline. The baseline layer is deterministic and SLM-oriented: it extracts text, reconstructs layout, segments the document, proposes candidates and maps them through reference tables. The LLM layer is deliberately constrained. It is used as a second reader for suspicious records, mixed-context cases and difficult semantic judgements, but it is not allowed to function as a free interpreter that invents unsupported corrections. This design reflects the thesis argument that auditability is more important than maximal generative flexibility. The detailed operational sequence is summarised in Appendix B.

Figure 1 summarises the operational logic of the pipeline. It should be read as a methodological map rather than as a claim that each step is fully automated or error-free.

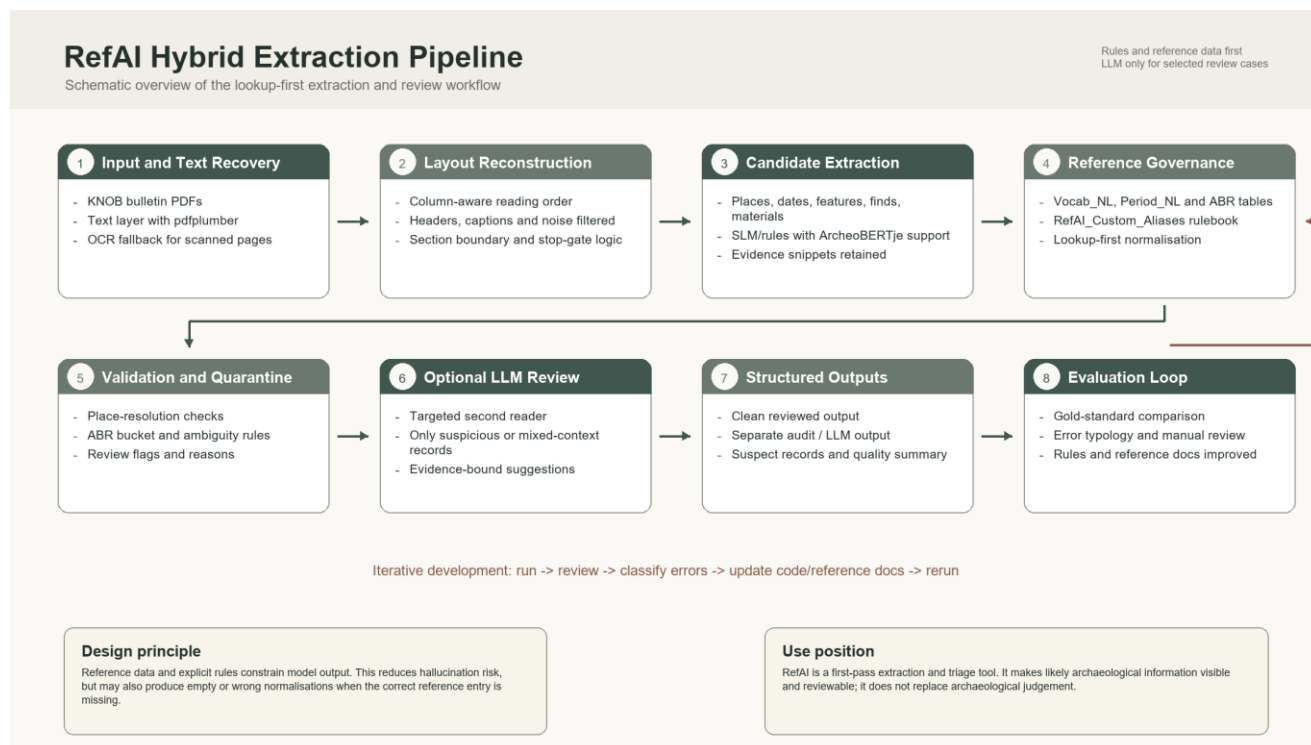


Figure 1. Schematic overview of the RefAI hybrid extraction pipeline, showing the lookup-first route from PDF input to structured output, review flags and iterative evaluation.

The workflow starts with PDF text extraction. When a text layer is available, pdfplumber-style extraction is attempted first. When the text layer is weak or absent, OCR (Optical Character Recognition) is used as fallback. OCR makes scanned pages machine-readable, but it also introduces errors in spelling, word boundaries and reading order. Layout reconstruction therefore becomes a methodological step rather than a technical detail: column order, captions, running headers and section boundaries affect what the later extraction layers can see. Document-AI research similarly shows that textual content cannot always be understood independently of spatial layout, especially in scanned and visually complex documents (Xu et al., 2020). Comparable work on retro-digitised historical address directories demonstrates that information extraction from older sources depends on an explicit processing pipeline for OCR-derived text, structural cues and entity normalisation rather than on raw keyword search alone (Khemakhem et al., 2018).

After text extraction, RefAI divides the text into sections and smaller chunks while retaining page ranges and nearby evidence. It then applies several rule families: layout and section-boundary rules, place and parent-settlement rules, dating patterns, ABR bucket rules, ambiguity and quarantine rules, and safeguards for optional LLM corrections. These rules determine what text is available to later stages, which candidates may be normalised and which cases must remain open for review. Appendix H provides a functional inventory of the principal rules, models and software components, including their purposes and limitations.

This layered methodology has a close parallel in the EITAB/Chronocarto project described by Mélanie-Becquet et al. (2020). That project also treats document segmentation, terminology, thesaurus construction and structured database enrichment as connected problems. Its observation that terminology and classifications change over time influenced RefAI's decision to use editable reference resources and explicit normalisation rather than treating extracted labels as timeless facts. In RefAI, the practical consequence was an iterative cycle of run, manual review, diagnosis, targeted change and renewed testing. The article is used as methodological support rather than as an accuracy benchmark, because it mainly documents the construction of an extraction framework.

This cycle was also a form of research-software development. RefAI was not produced through one final coding phase, but through small, inspectable changes followed by new runs on a fixed set of documents. Wilson et al. (2014) recommend precisely this kind of incremental development, together with meaningful names, modular functions, assertions, testing and version control, because scientific code must remain understandable after the first successful result. Wilson et al. (2017) similarly argue for 'good enough' practices that researchers can realistically maintain: explicit dependencies, organised project folders, explanatory documentation, examples and backups. These principles informed the later stabilisation of RefAI, including the separation of reference rules from procedural code, dated output folders, preserved backups and a documented startup route. They do not imply that the software meets every standard of a professionally maintained Python package. Rather, they provide a defensible framework for moving an experimental notebook toward a reproducible research object.

The pipeline is deliberately layered so that different kinds of failure can be distinguished. An OCR error is not the same as an incorrect ABR mapping, and an uncertain period is not the same as a wrong place. This separation shaped the trial-and-error methodology: after each run, errors could be assigned to a processing layer, followed by a targeted change and another controlled run. A single all-purpose model might produce cleaner-looking output, but it would also create a much larger black box whose errors could not easily be located or reviewed for accuracy. RefAI therefore keeps the usable output separate from logs, suspect records and LLM audit outputs. The result is more complex than one final spreadsheet, but it preserves the evidence needed to understand how that spreadsheet was produced.

4.1 ArcheoBERTje as Domain-Specific Semantic Support

A further methodological choice in RefAI is the use of ArcheoBERTje as a domain-specific language model within the local extraction layer. ArcheoBERTje is a Dutch BERT model for the archaeology domain developed by Alex Brandsen. According to the public model documentation, it is based on the Dutch BERTje model and was further fine-tuned on roughly 60,000 Dutch excavation reports from the DANS archive, representing approximately 650 million tokens. The accompanying GitHub repository also describes an archaeological NER version aimed at entities such as time periods, places, artefacts, contexts, materials and species. This makes the model especially relevant for RefAI, because the source material consists of Dutch archaeological news bulletin text rather than modern general-language prose. The associated research demonstrates that domain-specific pre-training improves archaeological named-entity recognition compared with generic multilingual and Dutch-language models (Brandsen et al., 2021a).

In the pipeline, ArcheoBERTje is not used as a free generative system. It supports semantic matching, meaning that it helps compare words by their likely archaeological meaning rather than requiring identical spelling. For example, an older or OCR-damaged expression may still resemble a known

ceramic or feature concept even when it is not written exactly like the reference label. This is useful because historical texts contain spelling variation, old terminology and uneven phrasing. However, similarity is only used to propose a candidate: local reference documents, thresholds, rulebook logic and validation checks still determine whether the candidate can be retained.

This provides a middle position between brittle exact matching and a generative model that may invent a plausible but unsupported classification. RefAI follows the same caution expressed in entity-linking research: one model or matching method should not be treated as conclusive on its own (de Vries et al., 2019; Ruiz et al., 2015).

This contribution is important for the thesis because it strengthens the lookup-first design without turning the system into a black box. The model helps RefAI deal with domain language, but the pipeline still records provenance, review flags and evidence snippets so that uncertain cases remain auditable. In practical terms, ArcheoBERTje has been most valuable as a bridge between messy source text and controlled archaeological terminology: it improves recall where exact aliases are incomplete, while the ABR rulebook and quarantine logic protect against overconfident mapping. I would also like to explicitly thank Alex Brandsen for his advice and for making ArcheoBERTje openly available. The public availability of this model made it possible to test a domain-specific NLP component within this thesis without having to train a new archaeological language model from scratch.

The model and its implementation are publicly documented through the ArcheoBERTje GitHub repository and Hugging Face model card, supporting inspectability and reuse of the domain-specific component (Brandsen, n.d.-a, n.d.-b).

5. Reference Governance, ABR Logic and Output Design

RefAI does not depend on a single ABR table or on model interpretation alone. It uses a layered reference architecture stored in `Reference_Docs`, with the active pipeline reading the Excel workbooks in `Reference_Docs/Excel_Docs`. The parent folder also contains CSV representations, but the evaluated notebook version loads the XLSX resources. These include `Vocab_NL`, with approximately 22,871 geographic entries and alternative names; `Period_NL`, with 125 chronological definitions; separate ABR tables for periods, finds and ground traces, materials, complexes, geomorphology, land use and find-layer characteristics; and `RefAI_Custom_Aliases`, which functions as an editable project-specific rulebook. Together these files form a small, versioned knowledge layer between the source text and the final structured output.

In the workflow shown in Figure 1, this chapter corresponds mainly to the reference-governance and validation stages: the point at which candidate terms are forced to pass through controlled vocabularies, ABR tables, project-specific aliases and ambiguity rules before they can become stable output values.

The geographic and chronological resources are therefore as important as the ABR resources. Place resolution compares extracted candidates with known place names, alternative spellings, geographic feature types, province information and contextual signals. The output can distinguish a normalised settlement from an alternative site, street or toponym through `Place_Normalised`, `Alt_Place_Name` and `Place_Resolution_Type`. Dating is handled through `Period_NL`, `ABR_Period` and additional dating aliases, which connect historical expressions and abbreviations to controlled period labels, codes and date ranges. This makes decisions more reproducible than asking a model to infer a place or period freely from its general training knowledge.

ABR mapping remains a central component because it connects variable historical language to professional Dutch archaeological terminology. The pipeline must determine whether a term belongs to a feature or context bucket, a finding or object bucket, a material bucket or a period bucket. Waterput (well) and gracht (ditch) are usually features or traces, Pingsdorf and Paffrath describe ceramic traditions or wares, and keramiek (ceramics) is a broader material-level concept. `RefAI_Custom_Aliases` adds

synonyms, rejection terms, author-name filters, field-type overrides, place and dating aliases, ambiguity rules, generic-suppression rules and restrictions on automatic LLM corrections.

Sensitive terms such as ring, haar (hair), band, kern (core), spits (point) and rechthoekig (rectangular) are not accepted merely because they match a vocabulary entry; they require local textual evidence or are retained in quarantine for review. This explicit governance follows the broader principle that controlled vocabularies and documented mappings make heritage data more interoperable and auditable (Harpring, 2010; CIDOC CRM SIG, 2024; Mélanie-Becquet et al., 2020).

A major reason for building this reference-first layer was to reduce hallucination. A language model can produce a plausible normalisation that is not supported by the source, whereas RefAI normally requires a connection to a known entry, alias, code or rule. The pipeline can record whether a result came from an exact lookup, fuzzy matching, a custom alias, province context, context carry or later review.

Unsupported candidates can remain empty, unresolved or quarantined instead of being silently converted into confident data. For archaeological reuse, a traceable omission is often less damaging than a convincing but untraceable fabrication, because the former can be inspected and corrected.

This design also creates a closed-world limitation: the pipeline can only normalise what its resources contain and what its extraction stages can connect to those resources. A historical spelling, local toponym, object term or dating expression may be absent, while OCR damage may prevent an existing entry from matching. Strict matching then produces an empty value, whereas fuzzy matching may select the nearest but wrong reference. The Egypt example encountered during testing illustrates a deeper semantic problem. A text may state that an object resembles examples known from Egypt without identifying Egypt as the archaeological location of the report. A gazetteer can confirm that Egypt is a place, but cannot by itself determine whether the mention expresses provenance, analogy or the actual find location. Reference data constrains the possible answer; it does not remove the need to interpret the relationship expressed in the sentence.

The resulting trade-off cannot be reduced to a simple claim that either model inference or reference lookup is always superior. A permissive system can improve recall but risks unsupported values. A strict reference system improves auditability but increases missing values and may occasionally choose an incorrect nearby concept. RefAI therefore combines broad candidate detection with conservative validation, evidence snippets, provenance, review flags and quarantine. Empty fields are not automatically treated as system failure: they can be an honest indication that the available text and reference layer do not justify a reliable normalisation. Nevertheless, the balance between these errors remains partly empirical and should be assessed on additional, unseen KNOB bulletins.

The most promising future improvement is consequently not an unlimited expansion of hard-coded exceptions, but systematic data mining of a larger KNOB corpus. Recurring place names, historical spellings, site names, object terms, period expressions and contextual co-occurrences could be collected automatically, reviewed by domain specialists and then added to versioned reference tables with provenance. The reference layer would thereby develop into a maintained domain resource rather than a static dictionary. This approach preserves the main advantage of the current design: corrections remain visible, editable and reusable. The clean reviewed output can remain compact, while audit and review outputs retain the evidence needed to understand how each value was produced.

6. Role of the LLM

The LLM is treated as a targeted second reader rather than as the primary extraction engine. In the current deposited version of RefAI, the Anthropic model used for this layer is `claude-haiku-4-5-20251001`. This model choice should be understood within the scope of the thesis: the project is not a benchmark of OpenAI, Anthropic and Gemini, nor does it claim that one provider is universally better than another. The research question concerns whether a lookup-first, rulebook-supported workflow can extract useful structured archaeological data while keeping uncertainty visible. The LLM is therefore included only as a controlled review component for cases where the deterministic SLM/rulebook layer signals uncertainty. This constrained role responds to the well-documented tendency of LLMs to produce fluent but unsupported content when they are asked to operate beyond reliable evidence (Huang et al., 2023).

Figure 1 also makes clear why the LLM is placed after rule-based validation rather than at the start of the workflow. Its role is to review selected uncertain cases after the SLM/rulebook layer has already produced evidence-bound candidates and review signals.

Anthropic/Claude was selected for this review role after practical testing within the project, because its responses were sufficiently structured for evidence-based record review and because the Haiku tier offered a more manageable cost than repeatedly using a larger model. This is a project-specific engineering decision rather than a general claim that Anthropic is superior to other providers. The tasks assigned to the model are narrow: checking whether a place assignment is supported by the supplied evidence, identifying mixed context, reviewing dating specificity and assessing whether an ambiguous ABR term should remain in the final output. More powerful models might improve some edge cases, but broad LLM review quickly becomes financially difficult to justify. The pipeline therefore limits both how often the model is called and how much authority its answer receives.

The thesis does not attempt a provider benchmark. Model quality, pricing and availability change over time, and a comparison based on informal impressions would not provide a stable academic basis for selecting a permanent winner. The defensible methodological choice is architectural rather than brand-based: RefAI does not depend on an LLM-first workflow. The local pipeline, reference documents and ABR rulebook remain central, while the commercial LLM is optional and restricted to cases where semantic review adds clear value. This reduces cost, improves traceability and limits exposure to model hallucination and provider changes (Huang et al., 2023; National Institute of Standards and Technology, 2023). This also affects the ethical scope of the project. As the developer, I wanted the pipeline to be as uncontroversial, locally controllable and transparent as reasonably possible. Without turning this thesis into a political comparison of technology providers, the preferred direction is to keep data processing close to Dutch and European research practice and to limit unnecessary dependence on non-European infrastructure for data, security and long-term availability. No major commercial LLM provider fully resolves European digital-sovereignty concerns, including Anthropic. RefAI therefore keeps OCR, layout reconstruction, ArcheoBERTje, reference tables, ABR validation and quarantine logic local and runnable without an API key. Claude is an optional and replaceable review layer rather than the methodological core.

In practical terms, the LLM receives only supplied evidence: the evidence quote, relevant chunk text and the candidate fields produced by the local pipeline. It is not asked to freely invent missing archaeological information. Its corrections are constrained by confidence thresholds, rulebook compatibility and auto-apply guards. This is important because the LLM can reason about context, but it can also over-correct, infer too much from weak evidence or introduce plausible-sounding but unsupported normalisations. The pipeline therefore treats LLM output as review evidence rather than as automatically authoritative truth.

Supplying bounded external evidence follows the general logic of grounded or retrieval-augmented language processing, while the additional rulebook restrictions address the fact that grounding reduces but does not eliminate model error (Lewis et al., 2020).

The current evaluation confirms this limited but useful role. In the June run, the LLM reviewed 87 of 93 records, marked 65 as needing correction, auto-applied 24 corrections, and identified 3 noise records and 12 mixed-context records. These numbers show that the LLM is not merely decorative: it detects uncertainty that the deterministic layer cannot always resolve. At the same time, it remains expensive and non-deterministic. The thesis should therefore present the LLM as a targeted review layer that supports triage and interpretation, not as the core extraction method and not as a substitute for human archaeological judgement.

7. Output Design, Review Flags and Quarantine

The review and quarantine mechanism was evaluated against the gold-standard place profiles for 93 reviewed records. It produced 36 true-positive flags, 49 false-positive flags, 3 false negatives and 5 true negatives. Under this operational definition, flag precision is 42.4%, flag recall is 92.3% and flag F1 is 58.1%. Its purpose corresponds to human-in-the-loop system design: automate routine processing while directing limited human attention toward cases where model confidence or rule consistency is insufficient (Wu et al., 2021).

This output logic is represented in Figure 1 by the transition from validation and quarantine to structured outputs: uncertainty is not removed from the pipeline, but carried forward as review information that can be inspected by a human user.

These numbers should be interpreted as triage metrics, not as ordinary extraction accuracy. The high recall shows that the system catches most records that differ from the gold standard. The low precision shows that it also flags many records that are ultimately usable. This is not automatically a failure. For a triage-oriented tool, over-warning can be acceptable if the goal is to prevent silent errors. The remaining challenge is to avoid making the review workload unnecessarily large.

The false-negative cases are especially informative because they show where RefAI can still look confident while differing from the gold standard. In this run there were three such cases, all in Test 1985 around Nijmegen. They were not missed place extractions, but field-level differences inside an already recognised place context. This suggests that the remaining difficult cases are mainly semantically dense records rather than complete failures of geographical extraction. This result supports the decision to frame RefAI as a triage system. The system is already useful when it extracts a candidate record and tells the user which parts deserve attention. High recall in the flagging layer means that few problematic records pass entirely unnoticed. Low precision means that the user still has to inspect more records than strictly necessary, but this is preferable to silently presenting uncertain data as reliable. Future work could make the flags more selective, but for the thesis the current behaviour is defensible because it prioritises caution and auditability.

8. Evaluation and Gold Standard Comparison

The gold standard was created and corrected by the author, Tobias Verwoert, who holds a bachelor's degree in Archaeology and, at the time of writing, has approximately six years of archaeological education and experience. This provides relevant domain competence for judging the selected notices, but it does not make the annotation objective. Only one assessor produced the target files, so no inter-annotator agreement could be calculated. Another archaeologist could reasonably select different evidence, split notices differently, choose another level of chronological specificity or organise the output columns differently. The gold standard is therefore an expert-informed target interpretation for this thesis, not a universally correct truth table. The scoring protocol and its limitations are documented in Appendix C.

The feedback arrow in Figure 1 is important for understanding the evaluation method. The comparison with the gold standard was not treated as a final binary judgement only, but also as a way to identify which rulebook entries, reference mappings and safeguards needed further adjustment.

A strict row-by-row comparison would therefore give a misleading impression of quality. If RefAI splits one long archaeological notice into two records while the gold standard keeps it as one, a row-based metric would count many fields as wrong even when the relevant information has in fact been detected. The reverse can also happen: RefAI may merge information that the gold standard separates. The evaluation therefore uses the archaeological signal as the main unit of comparison. The central question is not whether RefAI produces the same row structure as the human-edited file, but whether it manages to surface the relevant places, periods, features, findings and materials somewhere in the reviewed output, and whether uncertain or problematic information remains visible for review. Historical-document NER is particularly sensitive to this issue because errors can originate in OCR, span boundaries, entity classification or the source document itself (Ehrmann et al., 2021).

For this reason, the comparison uses a graded scoring method. A score of 2 means that the gold-standard term is present in the RefAI output in a directly usable form. A score of 1 means that RefAI has detected the core signal, but that the result still needs interpretation or correction before it can be treated as final. Examples include a site name that is captured as an alternative place name rather than as the main normalised settlement, a broad period code where the gold standard uses a more specific sub-period, or a semantically related ABR term that points to the correct archaeological category but not to the exact preferred label. A score of 0 means that the relevant signal is missing or not usable. This scoring system is less absolute than a binary match, but it better reflects how the pipeline is meant to function: as a tool that extracts and organises archaeological information for review, not as an autonomous final authority. This graded approach does not remove subjectivity, but it preserves more information than forcing every interpretative disagreement into a binary correct/incorrect label; recent work on subjective NLP evaluation similarly argues that annotator disagreement can represent meaningful uncertainty (Khurana et al., 2024).

The same principle is used for extra RefAI terms. An extra term is not automatically treated as the same kind of error in every situation. If an unsupported term silently enters the clean output as if it were certain, it is a serious false positive. If the same term is detected but marked through a review flag, quarantine rule or uncertainty indicator, it is better understood as a triage case. This distinction is crucial for RefAI, because many difficult terms are genuinely ambiguous. Ring can be an artefact, but can also appear as part of another word or as contextual noise. Huizen can refer to the place Huizen or simply to houses. Valkhof can be a meaningful site name but may need to be analytically grouped under Nijmegen. A good extraction system should not only find easy terms; it should also avoid hiding uncertainty when the text is semantically unstable.

The graded comparison was applied to the seven selected test PDFs. These contain 82 gold-standard rows and 93 RefAI reviewed-output rows. Across the evaluated fields of place, dating, feature, finding and material, the gold-standard files contain 301 target terms. RefAI retrieves 284 of these as exact usable matches and 4 as partial or reviewable matches, while 13 terms remain missing under this graded scoring method. These figures are deliberately presented as counts rather than as a single accuracy score. They indicate how much of the expert-selected archaeological signal in this specific test set is surfaced by the pipeline in a usable or reviewable form, while still leaving room for semantic disagreement and manual judgement. They should not be read as a representative performance estimate for all KNOB bulletins.

The category-level results show where RefAI is strongest and where the remaining weaknesses are concentrated. Place detection performs well: 50 of 51 gold-standard place signals are retrieved in usable form. Material extraction is also stable, with 37 of 37 target material terms retrieved. Feature extraction is similarly strong, with 91 of 93 terms retrieved. Findings and dating remain more fragile. Findings reach 72 of 77 usable matches, while dating reaches 38 of 43 usable or reviewable matches. This pattern is methodologically plausible. Materials are often expressed through concrete vocabulary such as pottery, bronze, iron or stone, and features often map well once the ABR rulebook is tuned. Dating and findings require more interpretation. Periods may be implicit, broad, narrow, mixed with publication years or dependent on artefact typology. Finds can be described through fragments, vessel types, generic material terms, or historical wording that does not map cleanly onto a single ABR label.

The comparison also highlights the value of the triage layer. RefAI produces 17 extra terms relative to the gold standard, but 16 of these occur in reviewable or flagged contexts. This does not remove the need for manual checking, but it changes how these cases should be interpreted. The system is not simply polluting the final dataset with invisible noise; in most of these cases it is surfacing uncertain candidates in a way that can be inspected. This supports the thesis argument that RefAI is most defensible as an extraction and triage prototype. Its value lies in quickly making relevant archaeological information visible and structured, while still directing attention to cases where human judgement is needed.

The results should therefore be interpreted with caution but not dismissed. The test set is small and partly shaped by the development process: the pipeline was repeatedly run, reviewed and improved using these documents. This means that the measured results are likely to be more favourable than they would be on a completely new and unseen set of bulletins. If additional documents were added, performance would probably drop at first, because new layouts, new OCR problems and new semantic edge cases would appear. Further tuning could then raise the scores again, but that would also increase the risk of adapting the system too closely to the evaluation material. The figures in this chapter are therefore best understood as a measurement of the current prototype on its development-and-evaluation set, not as a general accuracy claim for all archaeological bulletin text. The gold standard itself is also interpretative. Another archaeologist might choose a different level of dating specificity, preserve a site name differently, or classify a feature/object boundary in another way. In short, the evaluation attempts to make semantics measurable, and that is inherently problematic. Still, the data are not arbitrary: there are real overlaps between what RefAI extracts and what the gold standard expects. The comparison is useful because it identifies the type of work RefAI can support. It is especially useful for extracting low-hanging fruit: places, common periods, recurring features, common materials and many artefact categories. It is less reliable when the text requires deeper archaeological interpretation, when layout damage disrupts the evidence, or when terminology is ambiguous.

The evaluation therefore supports a limited but meaningful conclusion. RefAI should not be presented as a finished automatic extraction model that replaces expert judgement. It should be presented as a transparent workflow that can pre-structure messy archaeological bulletin text, retrieve a large share of relevant information, and expose many uncertain cases for review.

The detailed supporting workbook, RefAI_Graded_Gold_Standard_Comparison_20260617.xlsx, contains the document-level row counts, category-level summaries, term-level graded matches and extra RefAI terms used for this assessment. The most important outcome is not a single percentage, but the balance between retrieved information, reviewable uncertainty and remaining missed signals. Supporting evaluation detail and the error typology are provided in Appendix C, while Appendix E illustrates the compared Gold Standard and Reviewed RefAI structures.

Field	Gold target terms	Exact usable	Reviewable / partial	Missing	Interpretation
Place	51	50	0	1	Strong signal detection, but site/settlement choices remain interpretative.
Dating	43	34	4	5	Most periods are surfaced, but specificity and implicit dating remain difficult.
Features	93	91	0	2	Very strong retrieval, though feature/object boundaries still need review.
Findings	77	72	0	5	Useful retrieval, but artefact terminology is semantically sensitive.
Materials	37	37	0	0	Stable in this test set, mainly because material terms are often explicit.

This table uses absolute counts because the fields are semantically interpretative. The purpose is to show whether RefAI surfaces relevant archaeological signals for use or review, not to present the output as mechanically or fully accurate.

9. Known Limitations and Remaining Risks

The pipeline is limited because it tries to formalise archaeological judgement into deterministic steps. This is methodologically useful because it makes decisions traceable, but it is also inherently fragile. Every rule improves one pattern while risking a regression elsewhere. A strict block on Ring would reduce false artefacts, but could also remove real rings. A broad context-carry rule can rescue missing place names, but may accidentally carry a previous location into a new notice. A site-parent map can normalise Valkhof to Nijmegen, but it also raises the question of whether the analytical unit should be the site, the street, the settlement or the municipality.

The main remaining risks are OCR/layout damage, segmentation leakage, non-target text such as museum news or monumentenzorg (heritage management), homonyms, site-parent ambiguity, dating specificity and ABR bucket conflicts. These risks are not all solvable through more code. Some require better reference data, some require human review, and some should be reported as limitations of the method. The thesis should therefore be honest: RefAI is useful because it accelerates extraction and exposes uncertainty, not because it eliminates interpretation. The variety of these failures is consistent with the wider literature on historical-document extraction, which stresses that noisy text, changing layouts and domain-specific language interact rather than producing one isolated error source (Ehrmann et al., 2021).

The code is also methodologically limited because it turns flexible human reading into explicit rules. No controlled runtime, cost-per-document or reviewer-time study was conducted, and the number and cost of LLM calls varied with the records selected for review. The claim that RefAI can accelerate review is therefore plausible from its pre-structured outputs but not yet empirically demonstrated as time savings. Philip Verhagen and Dries Daems receive documentation for reuse, but this is not equivalent to a controlled usability study with professional archaeologists. Future evaluation should measure completion time, correction effort and user agreement alongside extraction quality.

10. Code Stability

The frozen evaluation identifier is ``RefAi_v8_1_0_slm_layout_compound_extraction``. The recommended deposited entry point is ``Code/Refai_V7_1_1_Slm_Local_Dating.py``, with ``Code/Refai_V7.ipynb`` retained as the development notebook. The deposition snapshot is ``RefAI_Deposition_Package``, updated and successfully run on 16 June 2026. Stable means that startup, path resolution, output structure, reference loading and the optional LLM switch can be reproduced; it does not mean that every extraction is correct. The technical environment and remaining gaps are listed in Appendix F.

For code stability, Figure 1 is useful because it separates the pipeline into identifiable stages. This makes it easier to explain which parts are considered stable for deposition and which parts remain dependent on reference maintenance, OCR quality or expert review.

Python was selected because it provides a readable high-level language and a mature ecosystem for PDF processing, OCR orchestration, tabular data, spreadsheets and natural-language processing. The official Python 3.12 documentation is therefore included as the authoritative technical reference for language behaviour and standard-library functionality (Python Software Foundation, 2026). RefAI's code was gradually divided into functions and rule families so that layout extraction, place resolution, dating, ABR mapping, review logic and output writing could be inspected separately. This modularity limits the effect of a local change and makes failures easier to trace, which is consistent with recommendations for robust research software (Taschuk & Wilson, 2017).

JupyterLab was valuable during development because code, intermediate output and methodological notes could be inspected in the same working environment. It also creates a reproducibility risk: notebook cells can be executed out of order, hidden state can survive between experiments, and a visually complete notebook does not guarantee that another user can rerun it from a clean session. Rule et al. (2019) recommend clear computational order, documented dependencies and restart-and-run-all checks for shared notebooks. RefAI addresses part of this risk by retaining the notebook as development history while providing a standalone Python entry point, a `Start\_Here` guide and installation checks in the deposition package.

The code-stability claim remains deliberately limited. RefAI was repeatedly validated through end-to-end batch runs, syntax checks, targeted regression checks and manual comparison of outputs, but it does not contain a comprehensive automated unit and integration test suite. Exact package versions were not frozen from the beginning, and the current evaluation documents also influenced development. Consequently, the deposited version is a reproducible thesis prototype rather than production software. Further changes should be versioned, tested first against the fixed evaluation set and then assessed on unseen bulletins before replacing this research version.

11. Depositing

The deposition package contains the runnable script and notebook, seven test PDFs, reference resources, gold-standard and evaluation material, example output and practical documentation. Machine-specific paths and API credentials are excluded. The SLM-only route runs without a commercial account; LLM review requires a locally supplied Anthropic API key.

The deposition package follows the same logic as Figure 1: it must include not only the Python code, but also the PDFs, reference documents, output templates, example outputs and documentation needed to reconstruct the full workflow.

The structure of the package is methodological rather than merely administrative. Reproducible computational research requires more than publishing the final code: the computational sequence, representative inputs, reference resources, configuration, environment assumptions and interpretation of outputs must also be recoverable. Sandve et al. (2013) emphasise recording operations, retaining exact inputs and versions and avoiding undocumented manual transformations. Wilson et al. (2017) add the practical importance of README files, example data, backups and explicit dependencies. This is why the RefAI package preserves the rulebook workbooks, sample PDFs, evaluation files and user-oriented startup documentation alongside the script.

A repository release should also identify the exact software version that produced the reported results. Katz et al. (2019) describe software citation as requiring importance, credit, unique identification, persistence, accessibility and specificity. For RefAI, this means that a future GitHub or archival release should receive a clear release tag and preferably a persistent identifier, while the accountability document should point to that exact version rather than to a continually changing repository. The downloaded programming and reproducibility literature is stored in `Literature\_Sources/Pdf\_Sources`; it supports the documentation but is not a runtime dependency of RefAI.

Data and code availability statement: the prepared local package is the current reproducible research object. A permanent GitHub or archival repository URL and, where available, a DOI have not yet been assigned and must be inserted after deposition. Public availability does not automatically equal an open licence; the final deposit must preserve attribution and confirm redistribution conditions for each external resource. Appendix G records the current rights and credit position.

12. Conclusion

For clarity, the four sub-questions are presented before the main research question, reflecting the order in which they are answered below.

Sub-question 1 - Segmentation robustness: *How can rule-based section segmentation be made reliable across different KNOB bulletin layouts, and what is the impact of segmentation errors on downstream extraction?*

Sub-question 2 - Hybrid extraction and hallucination: *To what extent can transformer-based candidate detection, combined with deterministic lookup and ABR mapping tables, reduce hallucination compared with an LLM-first pipeline?*

Sub-question 3 - Reference data and maintenance: *Which reference datasets for places, periods, ABR features and objects, dating expressions and materials contribute to accuracy and auditability, and how can a feedback loop maintain and expand them systematically?*

Sub-question 4 - Uncertainty and professional validation: *How should uncertainty, provenance and evidence snippets be represented in the output so that professional users can validate results efficiently without rereading full PDFs?*

Main research question: *How can a controllable, low-hallucination extraction workflow be built for Dutch archaeological texts that reliably extracts place, province, dating, features and objects into reproducible structured data for professional use?*

The first sub-question concerns segmentation robustness. The project shows that no single text-extraction or heading rule is adequate across the selected publication years. Native PDF text, OCR, column reconstruction, running-header filtering and section-boundary gates must be combined. When segmentation fails, the error propagates through the pipeline: a place name may be incorrectly carried over into the next notice, museum or monument-care text may leak into the archaeological output, or evidence from separate columns may be merged. The impact is therefore potentially larger than a single field error. The thesis did not conduct a controlled experiment in which segmentation was independently disabled and downstream performance recalculated, so its measurable causal impact cannot be expressed as a reliable percentage. The evidence is instead diagnostic: the reviewed runs repeatedly show that layout and boundary mistakes generate clusters of place, dating and entity errors.

The second sub-question also requires a qualified answer. Transformer-supported candidate detection and deterministic lookup reduce the opportunity for unconstrained generation because final values normally have to connect to source text, a reference entry or an explicit rule. This makes unsupported output less likely and more traceable than an LLM-first design. It does not eliminate error. Exact matches can still be semantically wrong, fuzzy matching can select a nearby but incorrect entry, and a closed reference set can return an empty value when the correct term is absent. Moreover, no equivalent LLM-first baseline was run under identical conditions, so the amount of hallucination reduction cannot be quantified conclusively. The result is an architectural and qualitative finding rather than proof that the hybrid system statistically outperforms every LLM-first alternative.

The third sub-question concerns the value and maintenance of reference data. The geographic vocabulary is essential for place and province resolution; Period_NL and ABR_Period constrain dating; ABR tables support feature, finding, material and context classification; and RefAI_Custom_Aliases records project-specific synonyms, rejection terms, ambiguity rules and field overrides. Their greatest contribution is not simply higher match counts, but auditability: a user can inspect and amend the resource that produced a decision. The project did not perform an ablation study that removes each resource in turn, so it cannot rank their individual contribution numerically. The implemented feedback loop nevertheless provides a practical maintenance strategy: recurring unmatched or misclassified terms are identified during review, checked against their evidence and added to versioned tables rather

than buried in undocumented model behaviour. Future expansion should use systematic data mining of unseen bulletins and expert validation, not only corrections derived from the present test set.

For the fourth sub-question, uncertainty is most useful when it is attached to evidence and to the origin of a decision. RefAI therefore preserves raw and normalised values, evidence quotes, place-resolution types, review flags and reasons, quarantined ambiguous terms, and separate clean and audit-oriented outputs. This allows a reviewer to distinguish a direct lookup from context carry, fuzzy resolution or LLM intervention. The approach is useful but not “frictionless”: too many flags create review fatigue, while thresholds that are too strict can hide valid information in quarantine. The system improves validation efficiency only when flags remain selective and when the evidence window contains enough context for an archaeologist to decide.

The technical, ethical and practical conclusions are therefore inseparable. Technically, RefAI is feasible as a modular prototype but requires continued reference maintenance, testing on unseen documents and expert review before broader deployment and before it can be used with 90–100% confidence that the output is correct and reflects the intended information. Practically, its value lies in accelerating first-pass extraction rather than eliminating labour; quality control is shifted and focused, not removed. Ethically, this narrower claim is important. The commercial LLM introduces cost, external data processing and dependency on non-European infrastructure, so it remains optional and is not allowed to become the sole basis of a final record. Human responsibility, evidence preservation and the ability to run the local pipeline without an API are safeguards, but they do not solve all governance questions. Reference-first processing also has an ethical trade-off: it reduces opaque hallucination but can create systematic omissions when vocabularies underrepresent particular places, periods or object terms.

Taken together, the answers to the four sub-questions allow the main research question to be answered positively, but only within a clearly bounded definition of reliability. RefAI demonstrates that a controllable hybrid, lookup-first workflow can extract and structure relevant archaeological signals from heterogeneous KNOB bulletins while retaining evidence and review information. OCR fallback, layout reconstruction, rule-based segmentation, ArcheoBERTje-supported candidate detection, controlled reference data, ABR validation and optional LLM review are technically feasible as a combined workflow. The output is not sufficiently stable to function as an autonomous archaeological database generator. Its defensible role is pre-processing and triage: it reduces the amount of text that must be read from scratch, but the resulting records still require expert validation.

The safest practical use is therefore within a known collection. An archaeological institution/company could, for example, use RefAI to make a first-pass database of its own excavation reports containing find locations, periods, ABR categories and object information. Because the organisation already knows the source material, omissions and unexpected classifications are easier to recognise and correct. Using RefAI blindly on an unfamiliar corpus is not recommended: unknown layouts, terminology and editorial conventions may produce convincing-looking output whose errors are difficult to notice.

RefAI should therefore be understood as a research prototype, meaning a functioning system built to test a method rather than a finished product. Its triage role is to perform the first selection: it surfaces likely places, dates, features, finds and materials and directs doubtful cases to a human reviewer. The pipeline does not possess archaeological judgement of its own. Its main strength is that much of the route from source text to output can be followed through evidence quotes, named rules, reference tables and review records. Its main weakness is that performance on unseen document designs remains uncertain. The next step should therefore be an external evaluation on KNOB bulletins that did not shape development, using independent archaeological annotation and measuring both retrieved information and the human effort needed for correction.

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The bibliography follows an APA-style author-date system. The sources are grouped by their role in the accountability argument rather than presented as one undifferentiated list. This makes clear which publications support the archaeological and digital-archaeology context, which sources support NLP and document-processing choices, which sources frame responsible AI use, and which sources document software, reproducibility and technical practice.

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Appendices

The appendices contain supporting explanations, definitions, evaluation detail and technical inventories. They are separated from the main accountability argument so that the main text remains readable while future users can still inspect the terminology, workflow, rules, software components, evaluation protocol and rights position in greater detail.

Appendix A. Glossary of Key Terms

This glossary explains the principal archaeological, computational and evaluation terms used throughout the document. The definitions describe how the terms are used within RefAI and are not intended as exhaustive definitions for every archaeological or computer-science context.

Term	Explanation
ABR (Archeologisch Basis Register)	A Dutch archaeological terminology and classification framework used to standardise periods, features, finds, materials and related concepts. RefAI uses ABR-aligned tables to connect variable source language to reusable labels and codes.
ABR bucket	A broad semantic category used by the ABR validator, such as feature, finding, material, period, place-risk or ambiguous finding. Bucket rules prevent terms from being normalised into an incompatible output field.
Alias	An alternative spelling, historical form, abbreviation or synonym linked to a preferred reference entry. Aliases help the pipeline recognise variants without changing the preferred normalised label.
Archaeological feature / spoor	A non-portable trace or context such as a ditch, posthole, grave, wall or well. Features must be distinguished from portable finds because they belong to different analytical and ABR categories.
ArcheoBERTje	A Dutch BERT-based language model for the archaeology domain, developed by Alex Brandsen and publicly available through GitHub and Hugging Face. RefAI uses it as local semantic support for candidate matching, not as a free generative model.
Artefact or finding / vondst	A portable object or find category recovered or discussed in an archaeological context, such as pottery, a coin or a ring. RefAI separates findings from features and materials.
Auditability	The degree to which a user can reconstruct how an output value was produced. In RefAI this is supported through evidence snippets, provenance, reference tables, review reasons and separate audit outputs.
Candidate detection	The stage in which rules or a language model identify text spans that may represent places, periods, features, findings or materials. A candidate is a proposal and is not automatically accepted as a final value.
Closed-world limitation	The limitation that a lookup-first system can only normalise concepts represented in its reference resources or rules. This constrains hallucination but can produce empty or incorrectly matched values when the correct entry is absent.
Context carry	A controlled rule that transfers a strongly established place or other context into a following text chunk when the continuation does not repeat it. Overly broad context carry can incorrectly attach a previous location to a new notice.
Controlled vocabulary	A maintained list of preferred concepts, labels and relationships. Controlled vocabularies reduce spelling and terminology variation, but their coverage and category choices require expert review.
Dating / datering	The extraction and normalisation of chronological information, including centuries, date ranges and archaeological periods. RefAI preserves both the interpreted period and the textual evidence used for it.

Term	Explanation
Deposition	The preparation and archiving of the research code, documentation, reference resources, test data and evaluation material so that the project can be inspected and reused.
Deterministic rule	An explicit instruction that produces the same result from the same input, unlike an unconstrained generative response. Deterministic behaviour improves reproducibility but can remain brittle when language or layout varies.
Entity linking	The process of connecting a detected text mention to a specific entry in a reference resource, for example linking a spelling variant to a place in the gazetteer.
Evidence snippet	A short quotation or text window retained alongside an extracted value. It allows a reviewer to inspect the immediate basis for a place, period or ABR classification without reopening the complete source document.
F1 score	The harmonic mean of precision and recall. It provides a combined score, but can conceal important differences between missed information and extra noise.
FAIR principles	Principles stating that research data and objects should be Findable, Accessible, Interoperable and Reusable. They inform the documentation and deposition choices in RefAI.
Fuzzy matching	Approximate string comparison used when an exact lookup fails, for example because of an OCR error or spelling variation. It can improve recall but may also select a similar yet incorrect reference entry.
Gazetteer	A structured geographic reference resource containing place names, alternative names and administrative or spatial attributes. RefAI uses a gazetteer for place and province normalisation.
Gold standard	A manually reviewed representation used as a comparison point for evaluation. In this project it is interpretative rather than absolute because archaeologists may reasonably choose different record boundaries or levels of specificity.
Hallucination	A fluent or plausible model output that is not adequately supported by the source or reference data. RefAI reduces this risk through constrained lookups, evidence requirements and restricted LLM auto-application.
Human-in-the-loop	A workflow in which people retain responsibility for reviewing difficult machine decisions. RefAI is designed to support archaeological judgement, not remove it.
Layout reconstruction	The recovery of reading order and document structure from page coordinates, columns, headings and text blocks. It is essential when OCR or PDF extraction returns text in an incorrect sequence.
LLM (Large Language Model)	A generative language model capable of interpreting and producing text. RefAI uses an Anthropic Claude Haiku model only as an optional second reader for selected uncertain cases.
Material / materiaal	The substance from which a finding is made, such as ceramic, stone, metal, wood or bone. A material is not necessarily an object type and is therefore stored separately.
NER (Named Entity Recognition)	Automatic detection and classification of meaningful spans such as place names, periods, objects or materials. Historical OCR and domain-specific language make NER more difficult.
Normalisation	The conversion of a raw extracted expression into a consistent preferred label or code, for example converting a spelling variant into a standard place name or ABR term.
OCR (Optical Character Recognition)	Technology that converts text visible in scanned page images into machine-readable characters. OCR enables extraction from image-only PDFs but can introduce spelling, word-boundary and reading-order errors.

Term	Explanation
Overfitting	Excessive adaptation to the known test documents, causing apparently strong evaluation results that may not generalise to unseen bulletins.
pdfplumber	A Python library used to extract text and layout information from PDFs that contain a usable text layer. RefAI attempts this before applying OCR fallback.
Precision	The proportion of extracted items that are relevant or correct. High precision means that the output contains relatively little extra noise, but it does not show how much relevant information was missed.
Provenance	Information about where a value came from and how it was produced, such as exact lookup, fuzzy match, context rule, model suggestion or human review.
Quarantine	A holding state for a detected term that may be relevant but lacks enough evidence for final acceptance. Quarantine preserves the candidate for review without presenting it as confirmed data.
RAG (Retrieval-Augmented Generation)	An approach in which a generative model receives information retrieved from external sources. RefAI is related to this grounding principle, but relies mainly on deterministic local tables rather than unrestricted retrieval and generation.
Recall	The proportion of relevant information in the comparison material that the system manages to retrieve. High recall means fewer missed signals, but may be accompanied by additional noise.
Reference-first / lookup-first workflow	A design principle in which extracted candidates are matched against explicit local resources before model-generated interpretation is considered.
Review_Flag	A machine-generated indication that a record or field requires attention because of ambiguity, conflicting evidence, weak resolution or another known risk.
Rulebook	The editable collection of project-specific aliases, bucket rules, rejection terms, ambiguity requirements and auto-apply restrictions stored outside the main procedural code.
Segmentation	The division of extracted document text into meaningful sections and chunks. Incorrect segmentation can mix places, articles, columns or non-archaeological sections and thereby cause multiple downstream errors.
Semantic matching	Comparison based on meaning rather than identical spelling. It can recover related terms that exact lookup misses, but requires validation because semantic similarity does not guarantee archaeological equivalence.
SLM (Small Language Model)	In this project, the local and comparatively lightweight language-processing layer used for candidate detection and semantic support. It operates together with rules and reference data and can run without a commercial LLM API.
Stress-test sample	A deliberately varied set of documents selected to expose difficult layouts, OCR quality and semantic edge cases. It is useful for development but is not automatically statistically representative.
Suspect records	A quality-control output containing records selected for attention because review flags, conflicts or other risk indicators are present.
Toponym / toponiem	A named local landscape element, site, street or place-related feature. A toponym may be useful as an alternative location while not being the settlement used in Place_Normalised.
Triage tool	The prioritisation of extracted information into directly usable records, uncertain cases requiring review and rejected noise. RefAI is primarily positioned as an extraction and triage workflow.

Term	Explanation
Unseen data	Documents that did not influence rule development, reference expansion or parameter tuning. Testing on unseen data is necessary to assess generalisation beyond the current seven development-and-evaluation bulletins.
Vindplaats (archaeological site or find location)	The location associated with archaeological observations or finds. It may differ from a municipality, settlement, street, site name or broader toponym.
Black box	A system whose internal route from input to output is difficult to inspect. RefAI attempts to reduce the black box by retaining evidence, separating processing layers and recording rules, reference matches and LLM interventions.
Geodatabase	A structured database designed to store and analyse geographic information together with related attributes. RefAI outputs can form a reviewed intermediate dataset before transfer to such a system.
Hybrid pipeline	A workflow that combines several methods rather than relying on one model. RefAI combines rules, reference tables, OCR, a local archaeological language model and optional LLM review.

Appendix B. Detailed Workflow Notes

The workflow begins with input selection and text recovery. RefAI first attempts native PDF extraction because embedded text generally preserves spelling, punctuation and character order better than OCR and is faster to process. When usable text is absent or nearly empty, pages are rendered and passed to RapidOCR. OCR is therefore a fallback rather than a default: it can restore otherwise inaccessible scans, but it may also introduce substitutions, broken words and uncertain reading order.

Text recovery is followed by column-aware layout reconstruction and section segmentation. This order is deliberate because entity extraction cannot be reliable if sentences from separate columns are merged, or if museum news, captions or monument-care sections are treated as ordinary archaeological notices. Running headers, likely captions and boundary headings are filtered or used as stop signals, while page ranges and nearby evidence remain attached to each chunk. These controls do not make historical layout deterministic, but they reduce the chance that a later extraction error was actually caused by earlier structural mixing.

Candidate extraction is intentionally broad, after which normalisation becomes conservative. Places and periods are checked against gazetteers, parent maps, aliases and dating tables; archaeological features, finds and materials are mapped through ABR resources and the RefAI_Custom_Aliases rulebook. This lookup-first design was chosen to make decisions inspectable and to reduce unsupported generation. Alt_Place_Name preserves sites, streets and toponyms when Place_Normalised should remain the parent settlement. Ambiguous terms are quarantined or flagged when local evidence is insufficient, which preserves potentially relevant information without presenting it as certain.

Finally, validated records are written to clean reviewed outputs, while LLM proposals, raw processing details, suspect records and quality-control information remain in separate audit-oriented files. Each batch receives its own dated run directory and retains evidence quotes, page ranges, review reasons and completion markers. This separation keeps the main dataset usable while preserving the provenance needed to investigate errors. The LLM is only a targeted second reader for selected doubtful cases, rather than the primary generator, because this limits cost, external dependency and the risk of fluent but unsupported corrections.

Appendix C. Evaluation Detail and Error Typology

This appendix records how the comparison categories were applied, which validity threats remain, and why recurring examples cannot be solved through simple hard blocks.

C.1 Gold-standard authorship and scoring protocol

The gold standard was produced by Tobias Verwoert as the author and sole assessor. His Bachelor's degree in Archaeology and approximately six years of archaeological education provide domain competence, but the absence of a second annotator means that no inter-annotator reliability statistic is available.

Category	Operational meaning
Correct / exact usable	The relevant signal is present at the expected meaning and specificity and can be used without substantive correction.
Semantic match	Different wording or normalisation expresses the same archaeological concept and remains directly usable.
Partial / reviewable	The core signal is detected, but specificity, field placement, parent-place relation or ABR label requires review.
Missing	A gold-standard signal is absent or unusable in the RefAI output.
Extra	RefAI produces a signal not retained in the gold standard; its seriousness depends on evidence and review status.
Flagged uncertainty	A doubtful candidate is preserved through a flag or quarantine rather than silently accepted as certain.

C.2 Threats to validity

Validity dimension	Threat and consequence
Internal validity	The same seven documents informed development and evaluation, so iterative tuning may inflate apparent performance. No controlled ablation isolated every pipeline component.
External validity	Seven varied KNOB samples do not represent all bulletin years, layouts or Dutch archaeological literature. Performance on unseen documents may initially decline.
Construct validity	The comparison converts semantic archaeological judgement into categories and counts. Record boundaries, specificity and feature/finding distinctions remain interpretative.
Assessment reliability	The author created the gold standard alone. Domain training supports the judgement, but no second annotator or inter-annotator agreement test was available.
Practical validity	Runtime, reviewer time and cost savings were not measured systematically, so efficiency claims remain provisional.

C.3 Representative semantic error cases

Case	Ambiguity	Cause	Current treatment	Remaining risk
Huizen	The word may denote the Dutch settlement or plural houses/features.	Place/common-noun homonym.	Context-sensitive ambiguity handling and review.	A short or OCR-damaged sentence may still lack enough evidence.
Ring	A genuine artefact term, but also OCR noise or a substring of another word.	Ambiguous finding and substring collision.	Term-local evidence, whole-word matching and quarantine.	Real rings may be withheld when evidence is weak.
Egypte	Mentioned as a comparison or origin associated with an object, not necessarily as the find location.	Gazetteer hit without correct semantic relation.	Require explicit find-location context.	Reference data confirms that Egypt is a place but cannot alone interpret the relation.
Linge	River, dike or bank context appeared near an Asperen notice.	Landscape feature mistaken for settlement.	Landscape guard and context carry to Asperen.	Historical descriptions can still introduce genuine new locations.
Kerkhof	Can function as an archaeological feature and as a gazetteer name.	Feature/place competition.	Feature priority unless explicit place context is present.	Some genuine place uses require manual confirmation.
Valkhof	A meaningful site name within Nijmegen.	Site versus parent-settlement normalisation.	Store Nijmegen as main place and Valkhof as alternative/site name.	The preferred analytical unit depends on the later research question.

Appendix D. Assessment and Professional Relevance

The project should be evaluated both as a scientific report and as a digital product. As a scientific report, the key criteria are the clarity of the research question, justification of methods, critical treatment of data, awareness of limitations and quality of argumentation. As a digital product, the key criteria are reproducibility, documentation, output structure, usability and transparency. RefAI's professional relevance lies in accelerating access to structured archaeological information while preserving reviewability and uncertainty.

Main-text word count estimate (Chapters 1-12, excluding front matter, references and appendices): approximately 9,167 words.

Appendix E. Gold Standard and Reviewed RefAI Sample Rows

The tables below show compact examples from the Excel sheets used in the evaluation. They are included to make the output template more concrete for the reader. They are not the full evaluation dataset; the complete gold-standard and reviewed RefAI workbooks remain available as separate Excel files.

These rows illustrate how the manually corrected gold standard represents the intended target interpretation for selected records.

Gold standard sample rows

Sample	Place	Dating	Features	Findings	Materials	Review
Gold Test 1	Vriezenveen	Middeleeuwen laat	Nederzetting, onbepaald	Bijl	IJzer	1
Gold Test 2	Hoogkarspel	Bronstijd	Zavel; Greppel/sloot; Kuil	steel; Andenne; Kogelpot; Pingsdorf; Plank; Afval	Keramiek; Brons	2
Gold Test 3	Nijmegen	Romeinse tijd vroeg	Zand; Greppel/sloot; Gracht; Cultusplaats/heiligdom/tempel	Constructie; Aardewerk, onbepaald	IJzer; Hout/houtskool	3
Gold Test 6	Tirns	Middeleeuwen	Laagte; Kuil; Greppel/sloot; Waterput	Afval; Kogelpot; steengoed	Keramiek	2
Gold Test 7	Deventer	Nieuwe tijd	Zand; Paalgat/paalkuil; Muur(restant); Waterput; Fundering	steengoed	Keramiek	3

Reviewed RefAI sample rows

These rows illustrate the kind of structured output produced by RefAI before final human interpretation. The comparison between these samples should be read in combination with Chapter 7, where the evaluation is framed as signal detection and triage rather than exact row-level reproduction.

Sample	Place	Dating	Features	Findings	Materials	Review
Reviewed RefAI Test 1	Vriezenveen	Late Middeleeuwen	Greppel/sloot	Bijl	IJzer	1
Reviewed RefAI Test 2	Hoogkarspel	Bronstijd	Zavel; Greppel/sloot; Kuil	steel; Andenne; Kogelpot; Pingsdorf; Plank; Afval	Keramiek; Brons	2
Reviewed RefAI Test 3	Nijmegen	Romeinse tijd	Gracht; Greppel/sloot; Waterput; afvalkuil; Kuil	Houtskool; Vloer	IJzer; Hout/houtskool; Kalk(steen)	1
Reviewed RefAI Test 6	Tirns	Middeleeuwen	Laagte; Kuil; Greppel/sloot; Waterput	Afval; Kogelpot; steengoed	Keramiek	2
Reviewed RefAI Test 7	Deventer	Nieuwe tijd	Zand; Paalgat/paalkuil; Muur(restant); Waterput; Fundering	steengoed	Keramiek	3

Appendix F. Technical Environment and Reproduction

The table distinguishes recorded configuration from recommendations and unresolved reproducibility work. Exact package versions were not frozen in the current package and should therefore not be inferred.

The package includes official Python 3.12 manuals and selected open-access literature on scientific programming, notebook reproducibility, robust research software and software citation in ``Literature_Sources/Pdf_Sources``. A source register in that folder explains why each item is included. These documents support the methodological account; they are not imported by the Python code. The most important remaining technical gap is the absence of a fully pinned environment and comprehensive automated test suite. A future release should add a machine-readable dependency lock file and automated smoke and regression tests.

Component	Recorded configuration or status
Frozen pipeline identifier	<code>`RefAi_v8_1_0_slm_layout_compound_extraction`</code> .
Recommended script	<code>`Code/Refai_V7_1_1_Slm_Local_Dating.py`</code> ; development notebook: <code>`Code/Refai_V7.ipynb`</code> .
Package snapshot	<code>`RefAI_Deposition_Package`</code> ; example successful run dated 16 June 2026.
Operating environment	Developed and packaged for 64-bit Windows with PowerShell startup scripts.
Python	Python 3.12-compatible environment is indicated by the deposited CPython 3.12 cache. The exact final interpreter version should be recorded in the release manifest.
Core packages	pandas, openpyxl, pdfplumber, PyMuPDF, Pillow, numpy, requests, torch, transformers, rapidocr-onnxruntime and python-docx.
Package versions	Exact dependency versions were not pinned in the current package. A generated requirements lock file is required for stronger long-term reproducibility.
Local model	<code>`alexbrandsen/ArcheoBERTje`</code> , downloaded through Hugging Face Transformers when enabled.
Commercial model	<code>`claude-haiku-4-5-20251001`</code> , Anthropic API version <code>`2023-06-01`</code> ; optional.
Hardware	No formal minimum-hardware benchmark was performed. A modern 64-bit computer with at least 8 GB RAM is a practical recommendation; GPU acceleration is optional, not methodologically required.
SLM-only start	Open <code>`Start_Here`</code> , run <code>`2_Start_Refai.ps1`</code> with PowerShell and choose menu option 4 (SLM-only batch).
SLM + LLM start	Set <code>`ANTHROPIC_API_KEY`</code> locally, open <code>`Start_Here`</code> , run <code>`2_Start_Refai.ps1`</code> with PowerShell and choose menu option 5 (SLM + LLM batch).

Appendix G. Rights, Attribution and Availability

Open access, open data and open-source software are not interchangeable. The project uses publicly accessible and reusable resources, but each retains its own copyright, licence or terms. The final repository should include a licence inventory and preserve all credits.

Resource	Credit	Current position and required action
KNOB bulletin PDFs	KNOB and the original authors	Used as publicly accessible scholarly source material. Public access is not automatically an open redistribution licence; citation and deposit rights must be confirmed before uploading the PDFs.
ABR resources	Relevant Dutch archaeological standard owners and source institutions, including RCE/SIKB where applicable	Used for research normalisation. The exact edition, source and reuse conditions should accompany the final deposited tables.
GeoNames-derived vocabulary	GeoNames	GeoNames states that its database is available under Creative Commons Attribution 4.0; attribution must be retained (GeoNames, n.d.).
ArcheoBERTje	Alex Brandsen and contributors	The public GitHub repository identifies GPL-3.0 for the repository. The model and its sources must be credited and their applicable terms preserved.
BERTje and Python dependencies	Their respective authors and maintainers	Each package/model retains its own licence. Dependency notices should not be replaced by a single general 'open-source' claim.
Anthropic Claude API	Anthropic	A paid external service governed by Anthropic's terms; no model weights or API credentials are redistributed.

Repository availability: [https://github.com/Solovyev/RefAI_MSc_Thesis_Submission]. Persistent identifier/DOI: [TBA].

Appendix H. Pipeline Rules, Models and Software Components

This appendix gives a functional overview of the principal controls used by the deposited RefAI pipeline. It is not a line-by-line reproduction of the Python code or every regular expression. Its purpose is to show where decisions are made, what evidence they use and where limitations remain. The authoritative implementation is the deposited Python script and notebook, while editable domain decisions are stored primarily in the Excel reference resources, especially RefAI_Custom_Aliases.xlsx.

Several rule families are deliberately stored outside the Python logic. RefAI_Custom_Aliases.xlsx contains term aliases, dating aliases, place aliases, rejection terms, field-type overrides, ambiguous-term evidence rules, bucket preferences, generic suppression rules and LLM auto-apply guards. Official or broader reference workbooks provide ABR concepts, periods, vocabulary, materials, find distributions, ground traces and related archaeological categories. This division allows recurring domain corrections to be documented in editable tables rather than added only as hardcoded exceptions.

Together, these components implement the hybrid architecture. No individual tool is considered a source of archaeological truth. Reliability comes from combining them, retaining evidence and requiring human review where the available information does not support an unambiguous decision.

H.1 Principal Rule Families

Rule family	Implementation examples	Function	Important limitation
Input and dependency controls	Environment switches; ensure_archaeobertje_dependencies; ensure_ocr_dependencies	Selects SLM-only or LLM-assisted operation, verifies required components and stops before output if ArcheoBERTje or a requested LLM is unavailable, or if no records are produced.	Availability checks confirm software presence, not the archaeological quality of its output.
Native text and OCR fallback	pdfplumber; PyMuPDF; RapidOCR; OCR text-length thresholds	Uses an existing PDF text layer where possible and activates OCR when the page contains too little usable text.	OCR can restore access to scanned text but may alter spelling, punctuation, word boundaries and reading order.
Layout and noise filtering	clean_ocr_text; line_is_noisy; caption and bibliography patterns; running-header filters	Removes likely editorial noise and prevents captions, headers and bibliographic fragments from becoming ordinary archaeological records.	Historical layouts are inconsistent; aggressive filtering can also remove genuine evidence.
Section-boundary controls	start, stop and ignore heading patterns; candidate heading classification; museum-news and monument-care gates	Determines where Archaeological News begins, continues or ends, including column-aware transitions.	A missed boundary can mix several notices; an early boundary can discard relevant archaeology.
Place resolution	place rejection lists; SITE_PARENT_MAP; STREET_PARENT_MAP; landscape guards; homonym rules; context carry	Distinguishes settlements from streets, sites, landscape terms, people and common nouns, while retaining useful alternative place names.	Context can be too weak to distinguish cases such as Huizen, Kerkhof or Egypte reliably.
Dating resolution	DATE_PATTERNS; dating aliases; local evidence windows; false-positive patterns; specificity ranking	Recognises explicit periods, centuries and local date expressions and prefers supported, sufficiently specific labels.	Implicit dating and mixed-period descriptions remain interpretative and may be missed or simplified.
Candidate and semantic support	trigger phrases; candidate spans; compound-term handling; ArcheoBERTje similarity thresholds	Finds possible archaeological terms and proposes semantically related controlled labels when spelling is not exact.	Semantic similarity is not proof of identity and can select a nearby but incorrect concept.

Rule family	Implementation examples	Function	Important limitation
ABR bucket and field governance	field_type_overrides; bucket_rules; ABR feature, finding and material groups	Keeps features, findings and materials in compatible fields and links recognised concepts to ABR-aligned labels and codes.	A reference entry may be missing, overly broad or unsuitable for the intended analytical level.
Ambiguity, quarantine and suppression	ambiguous_term_rules; exact-token evidence; false-context patterns; generic_suppression_rules	Requires nearby evidence for sensitive terms such as Ring, Haar and Band, and suppresses generic ceramic labels when a specific type is already present.	Conservative quarantine reduces false positives but can withhold genuine finds when source evidence is brief.
Review and confidence controls	derive_uncertainty_and_review; Review_Flag; evidence quotes; suspect-record criteria	Makes doubtful records visible and records why a value requires human inspection.	Too many flags create review fatigue; thresholds that are too strict can conceal useful candidates.
Optional LLM safeguards	suspicious-record selection; confidence thresholds; rulebook compatibility; auto_apply_guard_rules	Restricts Claude review to selected difficult cases and blocks unsupported automatic corrections.	A high-confidence LLM answer can still be wrong; commercial review also introduces cost and external processing.
Output and quality-control rules	clean reviewed output; raw and LLM audit outputs; suspect records; batch quality summary	Stores primary results in `Reviewed_Outputs_01` and LLM-reviewed results in `Llm_Review_Outputs_02`, while keeping quality-control, raw and diagnostic files separate. A complete batch writes `RUN_COMPLETED_SUCCESSFULLY.txt`; an interrupted or empty run retains `RUN_INCOMPLETE.txt`.	Multiple outputs improve accountability but require clear documentation so users know which file serves which purpose.

H.2 Models and Software Components

Component	Role in RefAI	Accountability consideration
pdfplumber	Attempts extraction from an existing PDF text layer.	Fast and direct, but dependent on whether the original PDF contains usable embedded text.
PyMuPDF, Pillow and NumPy	Render and process pages for OCR and layout-related operations.	Rendering choices such as resolution affect OCR quality and runtime.
RapidOCR / ONNX Runtime	Converts scanned page images into machine-readable text when native extraction fails.	OCR output must be treated as imperfect evidence rather than an exact transcription.
pandas and openpyxl-compatible Excel processing	Read reference workbooks and create structured CSV/XLSX outputs and quality-control tables.	Spreadsheet outputs preserve audit information but do not by themselves validate archaeological meaning.
PyTorch, Transformers and ArcheoBERTje	Provide local domain-specific semantic support for archaeological Dutch.	The model proposes similarity-based candidates; final acceptance remains constrained by evidence and references.
Anthropic Claude Haiku 4.5	Optional second reader for suspicious records and difficult contextual decisions.	Not required for SLM-only use; outputs are constrained, logged and never treated as automatically authoritative.
JupyterLab and PowerShell startup scripts	Provide an inspectable development notebook and a guided installation/run route for new users.	Reproducibility still depends on compatible software, available models and correctly preserved package structure.